Mat2002 Tutorial 1

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About me and this course I

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Quick review

- Check if a function is a solution to a given ODE
- Solve 1st order ODE
  - Normal form
  - Separable equation
  - linear equation

Check solution

Problem

Check $y = x + 3xln(x)$ *is a solution to* $x^2y'' - xy' + y = 0$

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Check $y = x + 3xln(x)$ *is a solution to* $x^2y'' - xy' + y = 0$

Solution

Consider $y = x + 3xln(x)$, *we have :*

$$y' = 1 + 3 + 3ln(x) = 4 + 3ln(x)$$
$$y'' = \frac{3}{x}$$

Hence $x^2y'' - xy' + y = 3x - 4x - 3xln(x) + x + 3xln(x) = 0$
Therefore,$y = x + 3xln(x)$ *is a solution to* $x^2y'' - xy' + y = 0$

Solve 1st order ODE

- Normal form: $\frac{dy}{dx} = f(x)$ or $y' = f(x)$

$$y = \int f(x)dx + C$$

- Separable equation $h(y)dy = g(x)dx$

integrate both side: $H(y) = G(x) + C$

Example

Solve $(1 + x)dy - ydx = 0$

Solution: We have $\frac{1}{y}dy = \frac{1}{1+x}dx$
Integrate both sides: $ln|y| = ln|x + 1| + C_1$
Thus $y = \pm e^{C_1}(1 + x)$
Therefore $y = c(1 + x)$

Losing a solution

Problem

- $\frac{dy}{dx} = y - 2$
- $x\frac{dy}{dx} - (y-1)^2 = 0$

Losing a solution

Problem

- $\frac{dy}{dx} = y - 2$

Solution

Constant function!

Q1: (i) $y = 2$ is a solution

(ii) $y \neq 2$

$$\frac{1}{y-2}dy = dx$$

$$\ln|y-2| = x + C$$

$$y - 2 = Ce^x$$

$$y = Ce^x + 2$$

Losing a solution

Problem

- $x\frac{dy}{dx} - (y-1)^2 = 0$

Solution

(i) $y = 1$ *is a constant solution*
(ii) If $y \neq 1$, *Then* $\frac{1}{(y-1)^2}dy = \frac{1}{x}dx$ *Integrate both side:*

$$-\frac{1}{y-1} = \ln|x| + C_1$$

$$y = \frac{1}{-\ln|x| - C_1} + 1$$

Remark

You can see in 1. The constant $C = 0$ *give us* $y = 2$. *But in 2. you can never get the solution* $y = 1$.

Linear equation

Linear equation: $a_1(x)y' + a_0(x)y = g(x)$.
Standard form: $y' + p(x)y = q(x)$
Integrate factor: $\mu = e^{\int p(x)dx}$.
We have $[\mu y]' = \mu(x)q(x)$

Example

Solve $y' - y = e^{3x}$

$$\mu = e^{\int -1dx} = e^{-x}$$

$$e^{-x} \cdot y' - e^{-x}y = e^{2x}$$

$$\left[e^{-x}y\right]' = e^{2x}$$

Integrate $e^{-x} \cdot y = \frac{1}{2}e^{2x} + c$

$$y = \frac{1}{2}e^{3x} + ce^{x}$$

Linear Equation

Problem

- *Solve* $x^3y' = -2x^2y + 1$.
- *Find the integrating factor for the differential equation and solve it,* $\cos(x)y' + \sin(x)y = 2x\cos^2(x)$.

Linear Equation

Problem

- *Solve* $x^3y' = -2x^2y + 1$.
- *Find the integrating factor for the differential equation and solve it,* $\cos(x)y' + \sin(x)y = 2x\cos^2(x)$.

Solution

Consider $x^3y' = -2x^2y + 1$. *we have:* $y' + \frac{2}{x}y = \frac{1}{x^3}$

Integrate factor: $\mu = e^{\int p(x)dx} = x^2$

Multiply it: $\left[x^2y\right]' = \frac{1}{x}$ *Integrate both side:*

$$x^2y = \ln|x| + C$$

$$y = \frac{\ln|x| + C}{x^2}$$

Linear Equation

Problem

- *Solve* $x^3y' = -2x^2y + 1$.
- *Find the integrating factor for the differential equation and solve it,* $\cos(x)y' + \sin(x)y = 2x\cos^2(x)$.

Solution

Consider $\cos(x)y' + \sin(x)y = 2x\cos^2(x)$
Which is: $y' + \tan(x)y = 2x\cos x$.
Integrating factor: $\mu = e^{\int \tan x} = \sec(x)$
Mutiply by μ : $\sec(x)y' + \tan(x)\sec(x)y = 2x$
Then $[\sec(x)y]' = 2x$
Integrate: $\sec(x)y = x^2 + C$
Then $y = C\cos x + x^2\cos x$.